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STORAGE SYSTEM IMPROVING ENCRYPTION INTEGRITY OF DATA AND OPERATING METHOD THEREOF

Abstract

A storage system may include a first device including a transmission circuit configured to generate encrypted first information by encrypting first information of a transmission target message, to generate encrypted second information by encrypting second information of the transmission target message, and to transmit the encrypted first information and the encrypted second information, and a second device including a reception circuit configured to decrypt the encrypted first information by receiving the encrypted first information and the encrypted second information from the first device and to store the first information that has been decrypted and the encrypted second information in a memory device.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATION

[0001] The present application claims priority under 35 U.S.C. § 119(a) to Korean Patent Application No. 10-2024-0027311, filed on Feb. 26, 2024, in the Korean Intellectual Property Office, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] Various embodiments relate to a computing system, and more particularly, to a storage system that improves the encryption integrity of data and an operating method thereof.

2. Related Art

[0003] In a computing system in which a plurality of computing devices constitutes a network, encryption technology is often used to ensure the confidentiality of messages that are transmitted and received between devices.

[0004] An integrity and data encryption (IDE) engine is an encryption device for protecting a message that is transmitted through a link of a device included in a computing system.

[0005] An object to be protected by IDE is a message transmitted through a link. Accordingly, in order to protect the message in a layer other than the link, a separate encryption scheme or device may be required.

[0006] There is a need for a method capable of protecting a message without influencing the power-performance-area (PPA) of a computing system because the PPA is affected when a separate encryption module is introduced.

SUMMARY

[0007] An embodiment of the present technology provides a storage system that improves the encryption integrity of data that have passed through a link layer by protecting the data without a separate encryption module, and an operating method of the storage system.

[0008] In an embodiment, a storage system may include a first device including a transmission circuit configured to encrypt a first information of a transmission target message, to encrypt a second information of the transmission target message, and to transmit the encrypted first information and the encrypted second information, and a second device including a reception circuit configured to receive the encrypted first information and the encrypted second information from the first device, and to decrypt the encrypted first information and to store the decrypted first information and the encrypted second information in a memory device.

[0009] In an embodiment, an operating method of a storage system including a first device and a second device may include generating, by the first device, encrypted first information by encrypting a first information of a transmission target message and generating encrypted second information by encrypting a second information of the transmission target message, transmitting, by the first device, the encrypted first information and the encrypted second information, decrypting, by the second device, the encrypted first information received from the first device, and storing, by the second device, the first information that has been decrypted and the encrypted second information in a memory device.

[0010] In an embodiment, a storage system may include a first device configured to generate encrypted first information by encrypting first information other than a data slot from a transmission target message including the data slot, to generate encrypted second information by encrypting second information including the data slot, and to transmit an encrypted message including the encrypted first information and the encrypted second information through a physical interface circuit.

[0011] In an embodiment, a storage system may include a second device configured to extract

encrypted first information from an encrypted message that has been received through a physical interface circuit, to generate first information by decrypting the encrypted first information using encryption information that has been received through the physical interface circuit, to extract encrypted second information from the encrypted message, and to store the first information that has been decrypted and the encrypted second information in a memory device.

[0012] According to the present technology, the PPA of the storage system can be improved by improving the confidentiality of data that have passed through a link without a separate encryption module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is a construction diagram of a storage system according to an embodiment of the disclosure.

[0014] FIG. 2 is a construction diagram of a storage system according to an embodiment of the disclosure.

[0015] FIG. 3 is a diagram for describing a message transmission and reception concept of a PCIe device.

[0016] FIG. 4 is a diagram for describing interfacing between PCIe devices according to an embodiment of the disclosure.

[0017] FIG. 5 is a diagram of a transmission circuit according to an embodiment of the disclosure.

[0018] FIG. 6 is a diagram for describing a method of transmitting a message according to an embodiment of the disclosure.

[0019] FIG. 7 is a diagram of a reception circuit according to an embodiment of the disclosure.

[0020] FIG. 8 is a diagram for describing a method of receiving a message according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0021] Hereinafter, embodiments of the present technology will be described in detail with reference to the accompanying drawings.

[0022] FIG. 1 is a construction diagram of a storage system according to an embodiment of the disclosure.

[0023] Referring to FIG. 1, a storage system **10** according to an embodiment may include an arbitrary number of first devices **102** and an arbitrary number of second devices **114**, which communicate with the first device **102** through peripheral component interconnect express (PCIe) or another data bus (not illustrated).

[0024] The first device **102** may be a host device or a data management device, for example. The second device **114** may be an external endpoint device, which may be, for example, a PCIe device, a data storage device or a server blade.

[0025] The first device **102** may include a root complex **104**, a processor **106**, memory **108**, an internal endpoint device **112**, and a physical interface **110**.

[0026] The root complex **104** may facilitate the formation of a network for the communication of data among the processor **106**, the memory **108**, an arbitrary number of internal endpoint devices **112** that are electrically coupled through the physical interface **110**, and the external endpoint device **114**.

[0027] For example, the root complex **104** may include a circuit, interfaces, or a program code for recognizing and communicating with devices electrically coupled to the storage system **10**, such as, the internal endpoint device **112** and the external endpoint device **114**. The root complex **104** may allocate resources to the devices or release resources allocated to the devices. In an embodiment, the root complex **104** may operate according to a data communication protocol, such as PCIe.

[0028] The internal endpoint device **112** and the external endpoint device **114** may communicate with the processor **106** and/or the memory **108** through the root complex **104**.

[0029] The processor **106** may include a circuit, interfaces, or a program code for controlling operations of components included in the storage system **10**. In an embodiment, the processor **106** may include a central processing unit (CPU), an advanced RISC machine (ARM), or an application specific integrated circuit (ASIC).

[0030] The memory **108** may store data, commands, and program codes that are necessary for an operation of the storage system **10**, and may include volatile and/or nonvolatile memory devices.

[0031] The memory **108** may store program codes to execute one or more operating systems (OSs) or virtual machines (VMs).

[0032] The processor **106** may execute the one or more OSs or VMs by driving program codes that have been stored in the memory **108**.

[0033] The physical interface **110** may be an adapter that electrically couples the internal endpoint device **112** and the external endpoint device **114** to the first device **102**. In an embodiment, the physical interface **110** may include a universal serial bus (USB) adapter, a peripheral component interconnect (PCI) adapter, a PCIe adapter, a small computer system interface (SCSI) adapter, a serial AT attachment (SATA) adapter, or a nonvolatile memory express (NVMe) adapter.

[0034] The internal or external endpoint device **112** or **114** may be a semiconductor device including an interface circuit that communicates with the root complex **104** according to a communication protocol of the root complex **104**, and may be a semiconductor device included in a storage device, for example.

[0035] FIG. **2** is a construction diagram of a storage system according to an embodiment of the disclosure.

[0036] Referring to FIG. **2**, a storage system **10-1** may include a root complex **104**, a processor **106**, memory **108**, at least one network interface **206**, a plurality of PCIe endpoint devices **210A**, **210B**, and **210C**, a PCI switch/PCIe bridge **212**, and a legacy endpoint device **214**. The PCI switch and the PCIe bridge may be separately constructed.

[0037] The plurality of PCIe endpoint devices **210B** and **210C** may each include an interface circuit that communicates with the root complex **104** through the PCI switch **212**. The legacy endpoint device **214** may communicate with the root complex **104** through the PCIe bridge **212**.

[0038] FIG. **3** is a diagram for describing a message transmission and reception concept of a PCIe device.

[0039] The root complex **104** and the endpoint devices **112**, **114**, **210A**, **210B**, and **210C** illustrated in FIG. **1** or **2** may each be denoted as a PCIe device, and may each communicate with another PCIe device through at least one interface model **30**, which is provided in each PCIe device.

[0040] For example, when a root complex **104** and endpoint devices **112**, **114**, **210A**, **210B**, and **210C** each perform communication based on a PCIe protocol, each interface model **30** may transmit and receive messages through a physical layer **310**, a data link layer **320**, and a transaction layer **330**.

[0041] Although not illustrated in FIG. **3**, the physical layer **310** may include a driver, an input and output buffer, a parallel to serial/serial to parallel conversion circuit, or an impedance matching circuit, and may convert a digital signal into an analog signal or convert an analog signal into a digital signal. When a message is transmitted, the physical layer **310** may serially convert messages that are received from the data link layer **320**, and may transmit a message by using a frequency and bandwidth that are compatible with another PCIe device that will receive the message. When a message is received, the physical layer **310** may transmit messages that are received from another PCIe device to the data link layer **320** by converting the messages in parallel.

[0042] The data link layer **320** may play a role as an intermediate step between the transaction layer **330** and the physical layer **310**. The data link layer **320** may encrypt a transmission target message, and may detect and correct an error of a received message. That is, link management and

message integrity may be implemented through the data link layer **320**.

[0043] The transaction layer **330** may provide the data link layer **320** with a message to be transmitted by packetizing the message, or may combine message packets that are received from the data link layer **320**.

[0044] FIG. **4** is a diagram for describing an interface between PCIe devices according to an embodiment of the disclosure.

[0045] Referring to FIG. **4**, a first device **410** may include a transmission circuit **411** and a reception circuit **413**. A second device **420** may include a transmission circuit **421** and a reception circuit **423**.

[0046] In an embodiment, the first device **410** may be a host device. The first device **410** may be a data management device, for example. The second device **420** may be an endpoint device. The second device **420** may be a PCIe device, for example, a data storage device or a server blade.

[0047] The transmission circuit **411**, **421** may separate a transmission target message into additional information, first information, and second information in order to transmit the message to another PCIe device. The message may be separated into a plurality of flits. Each flit may include a flit header (FLIT HEADER) and a plurality of slots. A slot may include an arbitrary number of header slots (HEADER SLOTS), an arbitrary number of generic slots (GENERIC SLOTS), and an arbitrary number of data slots (DATA SLOTS). In an embodiment, the first information may include a header slot (HEADER SLOT) and a generic slot (GENERIC SLOT). The second information may include a data slot (DATA SLOT). The additional information may include a flit header (FLIT HEADER).

[0048] The transmission circuit **411** may generate and encrypt a protection code for the first information, and may also encrypt the second information.

[0049] In an embodiment, the transmission circuits **411**, **421** may transmit, to respective reception circuits, encryption information (e.g., an encryption key, a counter value, or an initiation vector) that is used to encrypt the first information, but might not transmit, to the reception circuit, encryption information (e.g., an encryption key, a counter value, or an initiation vector) that is used to encrypt the second information.

[0050] The transmission circuits **411**, **421** may generate a first message authentication code (MAC) for an encrypted message in which the additional information, encrypted first information, and encrypted second information of a message to be transmitted have been combined.

[0051] The encrypted message and the first MAC may be transmitted to another PCIe device through a physical interface circuit.

[0052] The reception circuits **413**, **423** may receive the encrypted message and the first MAC from another PCIe device through the physical interface circuit.

[0053] The reception circuits **413**, **423** may generate a second MAC from the encrypted message, and may check the validity of the received first MAC by comparing the generated second MAC with the received first MAC.

[0054] The reception circuits **413**, **423** may separate an encrypted message into the additional information, the encrypted first information, and the encrypted second information, may decrypt the encrypted first information, and may check the integrity of the encrypted message.

[0055] The reception circuits **413**, **423** may use encryption information (e.g., an encryption key, a counter value, or an initiation vector) that is received from the transmission side in order to decrypt the first information. The reception circuits **413**, **423** will not decrypt the encrypted second information because encryption information (e.g., an encryption key, a counter value, or an initiation vector) of the second information is not provided by the respective transmission circuit.

[0056] The reception circuits **413**, **423** may store the additional information, the first information that has been decrypted, and the encrypted second information in a memory device.

[0057] In response to a read request for the second information, the transmission circuits **411**, **413** may generate a protection code by reading the first information associated with the second

information from the memory device, and may generate the encrypted first information by generating and encrypting the protection code.

[0058] The transmission circuits **411**, **421** may read, from the memory device, the additional information that is associated with the second information, and the second information that has been stored in an encrypted state, and may generate a first MAC for an encrypted message in which the additional information, the encrypted first information, and the encrypted second information have been combined.

[0059] The encrypted message and the first MAC may be transmitted to a PCIe device that has requested to read the second information through the physical interface circuit.

[0060] There is a need for a method capable of improving the confidentiality of data other than using the data link layer because the IDE engine only addresses the integrity of a message in the data link layer.

[0061] In the disclosed technology, data that are encrypted and transmitted by a transmission side may be stored in a memory device in an encrypted state without decrypting the data. The confidentiality of data can be improved by preventing the leakage of encryption information (e.g., an encryption key, a counter value, or an initiation vector) that is used to encrypt the data because the encryption information is not provided to the reception side and is managed on the transmission side.

[0062] FIG. 5 is a construction diagram of a transmission circuit according to an embodiment of the disclosure. FIG. 6 is a conceptual diagram for describing a method of transmitting a message according to an embodiment of the disclosure.

[0063] Referring to FIG. 5, transmission circuits **411**, **421** may include a message separation circuit **431**, a protection code generation circuit **433**, a first encryption circuit **435**, a second encryption circuit **437**, a message combination circuit **439**, and a MAC generation circuit **440**.

[0064] The message **431** may separate a transmission target message into additional information (ADD(ADDITIONAL AUTHENTICATION DATA)), first information (HEADER & GENERIC SLOT), and second information (DATA SLOT).

[0065] Referring to FIG. 6, a message may be separated into a plurality of flits. Each flit may include a flit header (FLIT HEADER) and a plurality of slots. The slots may include a header slot (HEADER SLOT), a generic slot (GENERIC SLOT), and a data slot (DATA SLOT). In an embodiment, the first information may include the header slot (HEADER SLOT) and the generic slot (GENERIC SLOT). The second information may include the data slot (DATA SLOT). The additional information may include the flit header (FLIT HEADER).

[0066] The protection code generation circuit **433** may generate a protection code (PCRC) for the first information (HEADER & GENERIC SLOT) including the header slot and the generic slot, and may combine plain text (P-PLAINTEXT_HG) of the first information and the protection code (PCRC).

[0067] The first encryption circuit **435** may encrypt the plain text (P-PLAINTEXT_HG) of the first information with which the protection code (PCRC) has been combined. In an embodiment, the first encryption circuit **435** may be an encryption circuit that uses an advanced encryption standard-counter (AES-CTR) mode. In this case, the first encryption circuit **435** may generate first information (C_HG, ECRC) that has been encrypted by encrypting the plain text (P-PLAINTEXT_HG) of the first information, with which the protection code (PCRC) has been combined, based on an encryption key (K), an initiation vector (IV), and a counter value (INC).

[0068] The encryption information (i.e., the encryption key (K), the initiation vector (IV), and the counter value (INC)) that is used to encrypt the first information may be transmitted from the transmission circuit to the reception circuit.

[0069] The second encryption circuit **437** may encrypt plain text (P-PLAINTEXT_D) of the second information including the data slot (DATA SLOT).

[0070] In an embodiment, the second encryption circuit **437** may be an encryption circuit using the

AES-CTR mode. In this case, the second encryption circuit **437** may generate second information (C_D) that has been encrypted by encrypting the plain text (P-PLAINTEXT_D) of the second information based on the encryption key (K), the initiation vector (IV), and the counter value (INC).

[0071] The encryption information (i.e., the encryption key (K), the initiation vector (IV), and the counter value (INC)) that is used to encrypt the second information might not be transmitted to the reception side.

[0072] The message combination circuit **439** may generate an encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)) by combining the additional information (ADD), the first information (C_HG, ECRC) that has been encrypted, the second information (C_D) that has been encrypted, and additional information (LEN(A), LEN(C)).

[0073] The MAC generation circuit **440** may generate a first MAC (MAC) based on the encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)). In an embodiment, the MAC generation circuit **440** may generate a hash value (GHASH(H)) from the encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)), and may encrypt the hash value (GHASH(H)) to generate an encrypted signal (AES-CTR(K)) based on the authentication key. Furthermore, the MAC generation circuit **440** may extract the most significant 96 bits (MSB.sub.96) of the encrypted signal as the first MAC (MAC).

[0074] The encrypted message (ADD/C_HG/C_D/ECRC/LEN (A)/LEN(C)) and the first MAC (MAC) that have been generated as described above may be transmitted to another PCIe device through the physical interface circuit.

[0075] FIG. 7 is a construction diagram of a reception circuit according to an embodiment of the disclosure. FIG. 8 is a conceptual diagram for describing a method of receiving a message according to an embodiment of the disclosure.

[0076] Referring to FIG. 7, reception circuits **413**, **423** according to an embodiment may include a MAC generation circuit **441**, a validity check circuit **443**, a message separation circuit **445**, a decryption circuit **447**, and an integrity check circuit **449**.

[0077] The MAC generation circuit **441** may generate a second MAC (MAC1) from an encrypted message (ADD/C_HG/C_D/ECRC/LEN (A)/LEN(C)) that is received through the physical interface circuit.

[0078] In an embodiment, the MAC generation circuit **441** may generate a hash value (GHASH(H)) from the encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)), and may encrypt the hash value (GHASH(H)) to generate an encrypted signal (AES-CTR(K)) signal based on the authentication key. Furthermore, the MAC generation circuit **441** may extract the most significant 96 bits (MSB.sub.96) of the encrypted signal as a second MAC (MAC1).

[0079] The validity check circuit **443** may check the validity of the encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)) by comparing the first MAC (MAC) that has been received through the physical interface circuit and the second MAC (MAC1) that has been generated by the MAC generation circuit **441**. When the encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)) is not valid, a device including reception circuits **413**, **423** may correct an error of the received message or may request the re-transmission of the message from the transmission side.

[0080] The message separation circuit **445** may separate the encrypted message (ADD/C_HG/C_D/ECRC/LEN(A)/LEN(C)) into additional information (ADD), first information (C_HG, ECRC) that has been encrypted, and second information (C_D) that has been encrypted.

[0081] The decryption circuit **447** may generate first information (HEADER & GENERIC SLOT) and a protection code (PCRC) by decrypting the first information (C_HG, ECRC) that has been encrypted.

[0082] In an embodiment, the decryption circuit **447** may be a decryption circuit using the AES-CTR mode. In this case, the decryption circuit **447** may decrypt the first information (C_HG,

ECRC) that has been encrypted based on an encryption key (K), an initiation vector (IV), and a counter value (INC) that have been transmitted by the transmission side.

[0083] The integrity check circuit **449** may check the integrity of the first information (HEADER & GENERIC SLOT) that has been decrypted based on the protection code (PCRC) that has been decrypted.

[0084] The additional information, the protection code (PCRC) that has been decrypted, the first information (HEADER & GENERIC SLOT) that has been decrypted, and the second information (C_D) that has been encrypted may be stored in a memory device. That is, the confidentiality of a data slot outside of the data link layer can be improved because the second information (C_D) that has been encrypted is not decrypted and is stored in the memory device in the encrypted state.

[0085] As described above, those skilled in the art to which the present technology pertains may understand that the present technology may be implemented in various other forms without departing from the technical spirit or essential characteristics of the present technology.

Accordingly, it is to be understood that the aforementioned embodiments are illustrative from all aspects not being limitative. The scope of the present technology is defined by the appended claims rather than by the detailed description, and all modifications or variations derived from the meanings and scope of the claims and equivalents thereof should be understood as being included in the scope of the present technology.

Claims

1. A storage system comprising: a first device comprising a transmission circuit configured to encrypt a first information of a transmission target message, to encrypt a second information of the transmission target message, and to transmit the encrypted first information and the encrypted second information; and a second device comprising a reception circuit configured to receive the encrypted first information and the encrypted second information from the first device, to decrypt the encrypted first information and to store the decrypted first information and the encrypted second information in a memory device.
2. The storage system of claim 1, wherein the transmission circuit transmits a first encryption information used to encrypt the first information to the second device, and does not transmit a second encryption information used to encrypt the second information to the second device.
3. The storage system of claim 1, wherein: the transmission target message comprises a plurality of flits, and each of the plurality of flits comprises a flit header, header slots, generic slots, and data slots.
4. The storage system of claim 3, wherein the first information comprises a header slot and a generic slot.
5. The storage system of claim 3, wherein the second information comprises a data slot.
6. The storage system of claim 1, wherein the transmission circuit is configured to separate the transmission target message into additional information, the first information, and the second information, generate a protection code from the first information and generate the encrypted first information by encrypting the first information and the protection code, generate a first message authentication code from an encrypted transmission target message in which the additional information, the encrypted first information, and the encrypted second information have been combined, and transmits the encrypted transmission target message and the first message authentication code to the second device.
7. The storage system of claim 6, wherein the reception circuit is configured to generate a second message authentication code from the encrypted transmission target message that has been received from the first device and check a validity of the encrypted transmission target message by comparing the second message authentication code with the first message authentication code, extract the encrypted first information from the encrypted transmission target message and check

an integrity of the encrypted transmission target message by generating the first information and the protection code by decrypting the encrypted first information, and store, in the memory device, the additional information and the encrypted second information that have been extracted from the encrypted transmission target message and the first information and the protection code that have been decrypted and generated.

8. An operating method of a storage system comprising a first device and a second device, the operating method comprising: generating, by the first device, encrypted first information by encrypting a first information of a transmission target message and generating encrypted second information by encrypting a second information of the transmission target message; transmitting, by the first device, the encrypted first information and the encrypted second information; decrypting, by the second device, the encrypted first information received from the first device; and storing, by the second device, the first information that has been decrypted and the encrypted second information in a memory device.

9. The operating method of claim 8, further comprising transmitting, by the first device, first encryption information that has been used to encrypt the first information to the second device, and not transmitting second encryption information that has been used to encrypt the second information to the second device.

10. The operating method of claim 8, wherein: the transmission target message comprises a plurality of flits, and each of the plurality of flits comprises a flit header, an arbitrary number of header slots, an arbitrary number of generic slots, and an arbitrary number of data slots.

11. The operating method of claim 10, wherein the first information comprises a header slot and a generic slot.

12. The operating method of claim 10, wherein the second information comprises a data slot.

13. The operating method of claim 8, further comprising: separating, by the first device, the transmission target message into additional information, the first information, and the second information, generating, by the first device, a protection code from the first information and generating the encrypted first information by encrypting the first information and the protection code, generating, by the first device, a first message authentication code from an encrypted message in which the additional information, the encrypted first information, and the encrypted second information have been combined, and transmitting, by the first device, the encrypted message and the first message authentication code to the second device.

14. The operating method of claim 13, further comprising: generating, by the second device, a second message authentication code from the encrypted message that has been received from the first device and checking a validity of the encrypted message by comparing the second message authentication code with the first message authentication code, extracting, by the second device, the encrypted first information from the encrypted message and checking an integrity of the encrypted message by generating the first information and the protection code by decrypting the encrypted first information, and storing, by the second device, the additional information and the encrypted second information that have been extracted from the encrypted message and the first information and the protection code that have been decrypted and generated in the memory device.

15. A storage system comprising: a device comprising a circuit configured to generate encrypted first information by encrypting first information other than a data slot from a transmission target message comprising the data slot, to generate encrypted second information by encrypting second information comprising the data slot, and to transmit an encrypted message comprising the encrypted first information and the encrypted second information through a physical interface circuit.

16. The storage system of claim 15, wherein the device generates a protection code from the first information, and generates the encrypted first information by encrypting the first information and the protection code.

17. The storage system of claim 15, wherein the device generates a first message authentication

code from the encrypted message comprising the encrypted first information and the encrypted second information, and transmits the encrypted message and the first message authentication code through the physical interface circuit.

18. A storage system comprising: a device comprising a circuit configured to extract encrypted first information from an encrypted message that has been received through a physical interface circuit, to generate first information by decrypting the encrypted first information using encryption information that has been received through the physical interface circuit, to extract encrypted second information from the encrypted message, and to store the first information that has been decrypted and the encrypted second information in a memory device.

19. The storage system of claim 18, wherein the device generates a protection code that is associated with the first information by decrypting the encrypted first information.

20. The storage system of claim 18, wherein the device receives a first message authentication code through the physical interface circuit, and compares a second message authentication code that has been generated from the encrypted message with the first message authentication code.
